

### 1. Your feature branch is behind main. How would you use git rebase to update your branch?

**A:** You use **git rebase** to move your feature branch's changes to the **end** of the main branch's history.

1. First, switch to your feature branch: `git checkout feature-branch`.
2. Then, run the rebase command: **git rebase main**. This command effectively rewrites the history, making it look like you started your work after all the new changes on main were completed.

### 2. A recently deployed commit caused issues. How do you remove the last commit and ensure the change is gone from history as well?

**A:** You use **git reset --hard HEAD~1**. This command moves the branch pointer back one commit and **permanently discards** the files/changes from that last bad commit in your local branch. Since it rewrites history, you must use `git push --force` to update the remote repository (use caution when forcing a push to shared branches).

### 3. How do you combine the last three small commits on your branch into a single, clean commit before merging?

**A:** You use **git rebase -i HEAD~3** (interactive rebase). This command opens a text editor where you change the action from pick to **squash** for the two older commits, keeping only the most recent one as the main commit. You then write a new, consolidated commit message.

### 4. You made a small typo fix in a file, but you forgot to include it in your last commit. How do you add that fix to the previous commit without creating a new one?

**A:** You stage the typo fix (`git add file-with-fix`). Then, you run the command **git commit --amend --no-edit**. This merges your staged fix into the previous commit, updating the commit message and hash ID.

**5. You need to download the latest changes from the remote repository but want to review them first before applying them to your current work. Which command do you use?**

**A:** You use **git fetch**. This downloads all the new commits and branches from the remote to your local repository. However, it keeps them separate from your working branch, allowing you to inspect the changes before merging or rebasing them.

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